FORMULA 1: Race Intelligence Dashboard - Final Report

1. INTRODUCTION

1.1 Project Overview

This data analytics project analyzes Formula 1 race performance, reliability patterns, and driver/constructor trajectories across the 2010–2026 seasons. It is designed to deliver actionable strategic insights for technical teams, motorsport analysts, and race management by transforming complex telemetry into an accessible visual format.

1.2 Purpose

The primary purpose of this project is to construct a fully interactive, web-based intelligence platform. By centralizing multi-season F1 data, it enables users to seamlessly track historical mechanical failures, evaluate collision rates, and monitor championship trajectories, thereby facilitating data-driven strategic decisions during race weekends.

2. IDEATION PHASE

2.1 Problem Statement

Inconsistent race telemetry tracking and fragmented DNF (Did Not Finish) data hinder real-time strategic decision-making and mechanical risk mitigation across multi-season F1 campaigns. Technical teams lack a consolidated interface to audit past performance against current race conditions.

2.2 Empathy Map Canvas

- **Users:** Race Strategists, Technical Directors, Motorsport Data Analysts.
- **Needs:** Quick, visual identification of mechanical vulnerabilities and constructor performance trends.
- **Pain Points:** Sifting through raw CSVs and disparate race result sheets takes too much time; lack of cross-filtering by specific circuits or eras.

2.3 Brainstorming

Initial ideas centered around creating static PDF reports. However, it was determined that an interactive Tableau dashboard, subsequently embedded within a custom Flask web application, would provide the necessary dynamic filtering (by Season and DNF Reason) that static reports

lack.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. **Data Ingestion:** Analyst accesses the raw dataset containing historical race results.
2. **Data Preparation:** Cleaning and standardizing geographic text and qualitative DNF reasons.
3. **Visualization:** Analyst interacts with the Tableau Public dashboard to uncover trends.
4. **Deployment:** End-user views the embedded insights via a local web browser application.

3.2 Solution Requirement

The solution requires a reliable dataset, a powerful visualization engine (Tableau), and a lightweight web framework (Flask) to host the dashboard locally without relying purely on proprietary software interfaces.

3.3 Data Flow Diagram

Raw CSV Data → Data Cleaning & Categorization → Tableau Public (Worksheets & Dashboards) → HTML/Iframe Embed Code → Flask Web Server → End-User Browser.

3.4 Technology Stack

Phase	**Technology / Tool**
Data Visualization	Tableau Public
Backend Routing	Python 3.13, Flask
Frontend	HTML5, CSS
Development Environment	Visual Studio Code, Windows Command Prompt

4. PROJECT DESIGN

4.1 Problem Solution Fit

The combination of a 5-chart dashboard and a 9-scene narrative embedded in a web application perfectly aligns with the problem statement, providing the interactivity needed to dissect fragmented data easily.

4.2 Proposed Solution

A unified "Formula 1 Race Intelligence Dashboard" featuring global circuit mapping, driver reliability analytics, and constructor championship trajectories, all operable via a user-friendly

local web app.

4.3 Solution Architecture

A local WSGI development server (`127.0.0.1:5000`) runs a Flask backend that serves an `index.html` template. This template contains a responsive iframe rendering the Tableau Public dashboard directly to the client.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- **Phase 1:** Data collection, geographic normalization, and DNF categorization.
- **Phase 2:** Tableau worksheet creation and interactive dashboard assembly.
- **Phase 3:** Structuring the 9-scene Tableau Story and publishing to Tableau Public.
- **Phase 4:** Python/Flask environment configuration, HTML embedding, and local testing.
- **Phase 5:** Project documentation formatting and final asset packaging.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The local Flask server successfully booted with zero native exceptions once the terminal environment was correctly configured. Cross-filtering latency within the embedded Tableau iframe was tested and found to perform seamlessly, inheriting Tableau Public's optimized cloud rendering speeds.

7. RESULTS

7.1 Output Screenshots

(Screenshots of the running localhost application and Tableau Public dashboards are compiled in the final project ZIP submission.)

8. ADVANTAGES & DISADVANTAGES

- **Advantages:** Highly interactive, centralizes years of fragmented data, easy to deploy locally, and visually intuitive for non-technical stakeholders.
- **Disadvantages:** The web app relies on an active internet connection to render the Tableau Public iframe; local server requires basic command-line knowledge to initialize.

9. CONCLUSION

The Formula 1 Race Intelligence Dashboard successfully resolves the issue of siloed race data. By integrating a meticulously cleaned dataset with Tableau's visualization capabilities and Python's web deployment frameworks, this project delivers a comprehensive, interactive tool ready for strategic race analysis.

10. FUTURE SCOPE

Future iterations could include live API integration for real-time race weekend telemetry (e.g., FastF1 API), predictive machine learning models to forecast DNF probabilities based on circuit weather, and a cloud-based deployment (e.g., Heroku or AWS) to remove the localhost requirement.

11. APPENDIX

Source Code

app.py:

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def dashboard():
    return render_template('index.html')

if __name__ == '__main__':
    app.run(debug=True)
```

Dataset Link

F1_Race_Intelligence_2010_2026.csv (Google Sheets format)

GitHub & Project Demo Link

Tableau Public URL: Formula 1 Narrative Story Dashboard
Project Demo: Hosted locally at http://127.0.0.1:5000 via Flask.